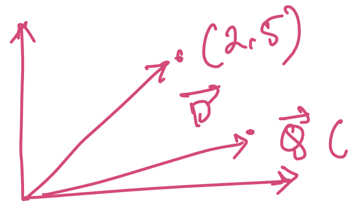
Support Vector Machine

Coordinate Geometry



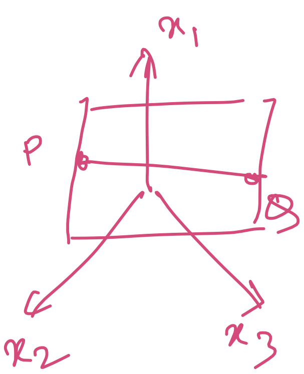
magnitude $= \sqrt{2^2+5^2} = \sqrt{29}$

$\vec{PQ} = \vec{Q} - \vec{P} = (5,5)$

$\vec{Q}\,(7,10)$

$\vec{p}\,(2,5)$

Angle $\tan\theta = y/x \Rightarrow \theta = \tan^{-1}(y/x)$



$f(x) = w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$

$w^Tx + w_0 = 0$

$w = \{w_1, w_2 \ldots w_3\}$

$x = \{x_1, x_2 \ldots x_3\}$

Normal vector of $f(x)$

$\vec{w} = [w_1, w_2, w_3] \Rightarrow \nabla f(x)$

show that $\vec{PQ}\cdot\vec{w} = 0$

$\vec{P} = w_1p_1 + w_2p_2 + w_3p_3 + w_0 \Rightarrow w^TP = -w_0$

$\vec{Q} = w_1q_1 + w_2q_2 + w_3q_3 + w_0$

$w^TQ = -w_0$

$w^TP - w^TQ = 0$

$\Rightarrow w^T(P-Q) = 0$

$= w^T\cdot\vec{QP} = 0$

Distance of a point z from the plane, $w^Tx + w_0 = 0$

$$d = \frac{w^Tz + w_0}{|w_0|} = \gamma = \text{geometric distance}$$

Functional distance $w^Tz + w_0 = \hat{\gamma}$

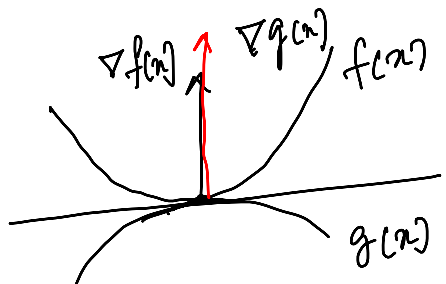
04/09/25 **Lagrange Multiplier**

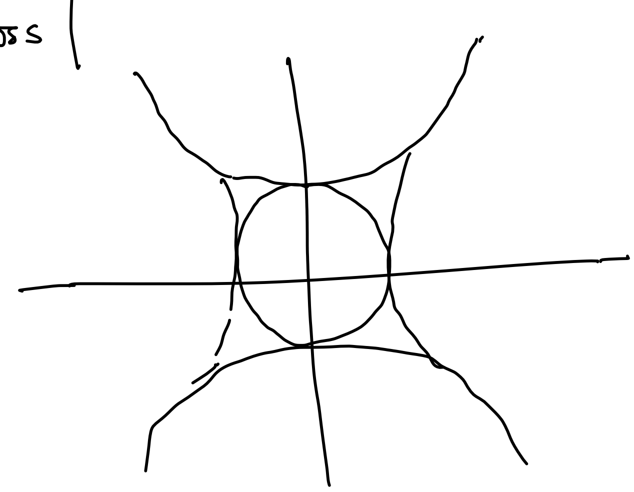
$\max_{x,y}\ z = x^2y$

s.t. $x^2 + y^2 = 1$

Regression – linear – Sq loss

– logistic – log loss





$$\boxed{\nabla f(x) = \lambda\, \nabla g(x)}$$

$\Rightarrow f(x) = x^2+y^2-4$ at $(2,0)$ Circle

$\nabla f(x) = (2x, 2y) = (4,0)$ at $(2,0)$

$g(x) = \frac{x^2}{4} + y^2 - 1$: Ellipse

$\nabla g(x) = (x/2, 2y)$

$(1,0)$

Lagrange Eq. $\mathcal{L}(x,\lambda) = f(x) - \lambda g(x)$ λ : langrange multiplier

max $f(x)$
s.t $g(x)$ equality

and $\nabla_x \mathcal{L} = 0$

$\nabla_\lambda \mathcal{L} = 0$

$f(x) - \lambda_1 g_1(x) - \lambda_2 g_2(x)$

$\nabla_x \mathcal{L} = 0$

$\nabla_{\lambda_i} \mathcal{L} = 0$

$\max_{h,s} f(h,s) = 200 h^{2/3} s^{1/3}$

s.t.c $20h + 170s = 20{,}000$

$\mathcal{L}(h,s,\lambda) = 200 h^{2/3} s^{1/3} - \lambda(20h + 170s - 20000)$

$$\left.\begin{aligned}
\frac{\partial \mathcal{L}}{\partial h} &= 200 \times \tfrac{2}{3} h^{-1/3} s^{1/3} - 20\lambda = 0 \\
\frac{\partial \mathcal{L}}{\partial s} &= 200 \times \tfrac{1}{3} h^{2/3} s^{-2/3} - 170\lambda = 0 \\
\frac{\delta \mathcal{L}}{\delta \lambda} &= -20h - 170s + 20000 = 0
\end{aligned}\right\}$$

$h = 666.66$
$s = 39.12$
$\lambda = 3.59$
$f(\ )_{max} = 51777$

KKT Theory
Karush Kuhn Tucker

max $f(x)$
s.t.c. $h_i(x) \le 0$

1. Convert it to LM take derivate w.r.t. variable and equate them to 0
2. $\boxed{\lambda_i h^i = 0}$ *
3. $h^i \le 0$
4. $\lambda_i \ge 0$

max: $-x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2$

s.t. $x_1 + x_2 \le 2$

$2x_1 + 3x_2 \le 12$

$x_1, x_2 \ge 0$

Cond^1: $\mathcal{L}(\ ) = -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2 - \lambda_1(x_1+x_2-2) - \lambda_2(2x_1 + 3x_2 - 12)$

$\frac{\partial \mathcal{L}}{\partial x_1} = -2x_1 + 4 - \lambda_1 - 2\lambda_2 = 0$ — (1a) ✓

$$\frac{\partial L}{\partial x_2} = -2x_2 + 6 - \lambda_1 - 3\lambda_2 = 0 \quad \text{— (1b)}$$

$$\frac{\partial L}{\partial x_3} = -2x_3 = 0 \Rightarrow x_3 = 0$$

Cond^n 2

$\lambda_1 (x_1 + x_2 - 2) = 0$ — (2a)

$\lambda_2 (2x_1 + 3x_2 - 12) = 0$ — (2b)

Cond 3

$x_1 + x_2 - 2 \le 0$ — (3a)

$2x_1 + 3x_2 - 12 \le 0$ — (3b)

Cond 4 : $\lambda_1, \lambda_2 \ge 0$

Case 1: $\lambda_1 = 0$ $\lambda_2 = 0$ (1a) (1b) $x_1 = 2$, $x_2 = 3$

X (3a) $3 \le 0$ X

Case 2 $\lambda_1 \ne 0$, $\lambda_2 \ne 0$; $\left.\begin{array}{l} x_1 + x_2 - 2 = 0 \\ 2x_1 + 3x_2 - 12 = 0 \end{array}\right\}$ $x_1 = 8$ $x_2 = -6$

(1a) (1b) $\lambda_2 = -26$ X

Case 3 $\lambda_1 = 0$, $\lambda_2 \ne 0$ (1a) (1b) $\left.\begin{array}{l} -2x_1 + 4 - 2\lambda_2 = 0 \\ -2x_2 + 6 - 3\lambda_2 = 0 \end{array}\right\}$ $x_1 = \frac{2}{3} x_2$

(2b) $2x_1 + 3x_2 - 12 = 0$

$\Rightarrow x_1 = 2$; $x_2 = 3$ (3a) $5 - 2 \le 0$ X

Case 4 $\lambda_1 \ne 0$ $\lambda_2 = 0$ $x_1 = 1/2$ $x_2 = 3/2$ $\lambda_2 = 0$ $\lambda_1 = 3$

Primal and Dual theorem

$\min_w f(w)$

w.r.t. $g_i(w) \le 0$ $i = 1 \dots k$

$h_i(w) = 0$ $i = 1 \dots l$

Generalized Lagrange Eq^n: $L(w, \alpha, \beta) = f(w) + \sum_{i=1}^{k} \alpha_i g_i(w) + \sum_{i=1}^{l} \beta_i h_i(w)$

Let us define: $\theta_P(w) = \max_{\substack{\alpha, \beta \\ \alpha \ge 0}} L(w, \alpha, \beta)$

$= \max_{\substack{\alpha, \beta \\ \alpha \ge 0}} f(w) + \sum \alpha_i g_i(w) + \sum \beta_i h_i(w)$

Cond^n Violated

$g_i(w) > 0$ $\alpha_i \to \infty$ $\theta_P(w) \to \infty$

$h_i(w) \ne 0$ $\beta_i \to +\infty / -\infty$ $\theta_P(w) \to \infty$

Cond^n Satisfied

$\theta_P(w) = f(w)$ | $\theta_P(w) = \begin{cases} f(w) & \text{Satisfied} \\ \infty & \text{Otherwise} \end{cases}$

Primal $\min_{W} \theta_p(w)$ | dual

Soln of the primal problem

$$p^* = \min_{w} \max_{\alpha,\beta} L(w,\alpha,\beta) \quad \Big| \quad d^* = \max_{\alpha,\beta} \min_{w} L(w,\alpha,\beta)$$

max-min inequality theory

$$\max \min f(x) \le \min \max f(x)$$

$$d^* \le p^*$$

$$d^* = p^*$$

① f and g should be convex and h should be affine

② g_i are (strictly) feasible $g_i(\,) \le 0 \quad \forall i$

there must exist w^*, α^*, β^* and $d^* = p^* = L(w^*, \hat{\alpha}, \hat{\beta})$

and w^*, α^*, β^* should satisfy KKT

$$\frac{\partial L}{\partial w} = 0 \qquad \frac{\partial L}{\partial \alpha} = 0 \qquad \frac{\partial L}{\partial \beta} = 0 \qquad \alpha_i^* g_i(\,) = 0$$

$$g_i(\,) \le 0$$

$$\alpha^* \ge 0$$

1960's Vladimir Vapnik & Alexey Chervonenkis

1961's $(x, y) \rightarrow$ Kernel $\rightarrow$ transform a non-linear data to a higher dimension with linear separability

$$K(x, y) = \cancel{\#}$$

1968 VC dimension : Complexity of the model w.r.t the data point $\Rightarrow$ how can a model be generalized to unknown data.

1990 AT&T bell lab

1992 $\boxed{\text{COLT}}$ $\leftarrow$ kernel-tricks

1995 $\rightarrow$ SVM $\rightarrow$ Soft-margin

98 – $\boxed{\text{SMO}}$ $\rightarrow$ Scikit learn